Recovery-Time Guarantees for Safety Filtering under Sampled Actuation

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***Abstract*— Safety alone can leave a vehicle at an excessive gap from a stopped leader. We study input-constrained longitudinal recovery under held actuation. Safety permits time-dependent dynamics; exact performance assumes autonomous dynamics behind a stopped leader. A phase-plane profile gives minimum entry time T^*, including launch failure and trapping. A feasible fraction $\theta \in (0,1)$ defines an energy floor and the exact worst-case clock W_θ shared by a specified continuous-time nominal-fidelity class. A constructive boundary-clearance certificate supplies constant-input holds through launch and preserves $T_{\mathrm{entry}} \le W_\theta \le T^*/\sqrt{\theta}$; exact sampled detection adds at most one period. For three models, periods $.01$ and $.025$ are certified analytically and by offline interval bounds, respectively, uniformly within 12 specified episode bands. All 192 experiments at these fixed periods satisfy the certificate. A progress-aware CBF is usually faster and cheaper in a separate $.05$ comparison; no hard real-time claim is made.**

***Index Terms*— Safety filters, sampled-data control, finite-time reachability, minimum-time control, nominal fidelity, viability theory.**

I. Introduction

A safety filter may prevent collision yet recover a desired gap slowly under an unfavorable nominal. Digital actuation also requires safety between updates. For input-constrained longitudinal dynamics, $\theta \in (0,1)$ specifies an energy floor relative to fastest recovery.

CBF–CLF programs address actuator-constrained cruise control [1], [2]; backup CBFs certify backup trajectories [3]. Fixed-time constrained control [4], [5], [6], timed reach–avoid [7], HJ safety/liveness [8], and faithful return/recurrence filters [9], [10] already combine safety and progress. Path CBFs also provide liveness and completion-time bounds [11]. We identify the exact common worst clock for a specified fidelity class relative to the same system's minimum entry time.

The spatial benchmark follows phase-plane minimum-time control [12], [13] and path parameterization [14], [15]; reflection uses a Skorokhod map [16]. We explicitly treat physical clocks, launch failure, and asymptotic trapping. Sampled reachability [17], CBF margins [18], approximate discrete models [19], self-triggering [20], and zero-order CBFs [21] address intersample safety. Input-delay certificates [22] and feasible mechanical constraints [23] cover broader implementation issues; sampled finite-time tasks [24] and early activation [25] also address timed progress. Our clearance budget uses this established margin principle. Its specialization couples ordered boundaries, a positive rest launch, and a ceiling-clock prefix to the exact recovery reference:

1) Whole-domain recoverability and an exact scalar minimum-entry benchmark, including launch and trapping (Theorems 2–3).
2) The same exact worst-case clock W_θ for every filter in the specified strict-fidelity class. Braking attains it; the factor $1/\sqrt{\theta}$ is sharp under stated rest-start conditions (Theorem 5).
3) A closed spatial held-input interval, constructive joint-boundary feasibility through launch, and useful complete-episode periods (Theorem 7, Proposition 8, Corollary 9). Holds preserve $T_{\mathrm{entry}} \le W_\theta$; exact sampled detection adds at most a period.

Full proofs, protocols and the result map are in the versioned technical notes [26].

II. Model and recovery scope

We first allow time-dependent dynamics and a moving leader:

$$\dot d = w(t) - v, \quad \dot v = a(t,d,v) + b(t,d,v)u, \quad |u| \le U. \tag{1}$$

Here d is the follower–leader gap, v the follower speed, and w the leader speed. The function $0 \le w \le M$ is continuous; a, b are continuous, locally state-Lipschitz and of at most linear growth uniformly on finite time intervals, and $b \ge b_* > 0$. Let $U, M, \gamma, \epsilon > 0$ and set

$$h = d - \gamma v - \epsilon, \qquad D = \{h \ge 0,\ 0 \le v \le M\}. \tag{2}$$

Write $A_\pm = a \pm bU$ and $p = \dot v \in [A_-, A_+]$. For every $t \ge 0$, exact viability on the lower-speed face $v = 0$, upper-speed face $v = M$, headway face $h = 0$, and their intersections requires

$$\begin{aligned} A_+ \ge 0\ (v=0), \qquad A_- \le 0\ (v=M), \\ A_- \le (w-v)/\gamma \quad (h=0). \end{aligned} \tag{3}$$

These are pointwise tangent conditions [27], [28]. For $h_g \ge 0$, $h_s > 0$, and $\lambda \ge 0$, put

$$\rho_g(h) = \frac{(h-h_g)_+}{h_s + (h-h_g)_+},$$

$$r_\lambda = \frac{w/2 - v + \lambda \rho_g(h)(M-v)}{\gamma}, \tag{4}$$

$$u_b = \mathrm{clip}_{[-U,U]}\, \frac{r_\lambda - a}{b}. \tag{5}$$

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Proposition 1 (Nonautonomous safety)**.** *Under* (3)*, the continuous-time feedback* (5) *preserves* D*.*

Proof. The reference satisfies $r_\lambda \geq 0$ on $v = 0$, $r_\lambda \leq 0$ on $v = M$, and $r_\lambda \leq (w - v)/\gamma$ on $h = 0$. Its projection onto $[A_-, A_+]$ retains all active inequalities by (3). Tangent-cone invariance applies; a complete nonautonomous proof is given in [26, Prop. 4.0]. □

Exact-results scope. Henceforth $w = 0$ and a, b are autonomous, locally Lipschitz, with at most linear growth; nominals may remain time dependent. This specialization covers recovery, optimality, fidelity, and sampled actuation. Proposition 1 alone gives no sampled guarantee. For $h_g \geq 0$, define

$$d_g = \epsilon + h_g, \qquad \mathcal{T}_g = \{(d, 0) : \epsilon \leq d \leq d_g\}. \tag{6}$$

For $\eta > 0$, the entry target is $d \leq d_g + \eta$, with free terminal speed. Every safe trajectory has

$$d \downarrow d_\infty \geq \epsilon, \qquad \int_0^\infty v\, dt = d_0 - d_\infty < \infty, \qquad v \to 0. \tag{7}$$

Bounded state and acceleration give uniform continuity, hence $v \to 0$.

Theorem 2 (Whole-domain recoverability)**.** *Under* (3)*, every state in* D *can be safely driven asymptotically to* $\mathcal{T}_g$ *iff*

$$A_+(d, 0) > 0 \qquad (d > d_g). \tag{8}$$

For any $h_s, \lambda > 0$*,* (5) *with* $w = 0$ *is a regular recovery witness.*

Proof. At an excess rest state with $A_+(d_*, 0) = 0$, write $q = d_* - d \geq 0$ and $Y = q + v$. Local Lipschitz continuity gives $\dot{Y} \leq (1 + L)Y$, $Y(0) = 0$, so every safe measurable input is trapped by Grönwall. Conversely r_λ satisfies all face tangent inequalities; acceleration projection retains them by (3). If its limiting gap exceeded d_g, both r_λ and A_+ would stay positive near the limit, contradicting $v \to 0$ in (7). □

This separates absent authority from nominal-induced stopping [26, Thm. 2.1]. Under (8), the witness has a uniform low-speed reserve on compact excess-gap intervals. Define $\tau_\eta = \inf\{t \geq 0 : d(t) \leq d_g + \eta\}$, with $\inf \varnothing = +\infty$. For $d_0 > d_g + \eta$, if $\dot{v} \geq c_\eta > 0$ on the pre-entry strip $d_g + \eta \leq d \leq d_0$, $0 \leq v \leq \nu_\eta$ for some $0 < \nu_\eta < M$, uniformly in time and over admissible nominals, then

$$\tau_\eta \leq \frac{(\nu_\eta - v_0)_+}{c_\eta} + \frac{d_0 - d_g - \eta}{\nu_\eta}. \tag{9}$$

An initially satisfied target has $\tau_\eta = 0$; the strip argument and regular-filter application are in [26, Thm. 3.1, Prop. 4.1]. We next compare recovery with minimum entry time under explicit nominal fidelity. For time-dependent coefficients, pointwise $A_+(t, d, 0) > 0$ is insufficient: with $U = b = \gamma = M = 1$ and $a = e^{-t} - 1 - v$, all face tests hold but a rest start satisfies $v(t) \leq te^{-t}$ and travels at most one unit. A time-uniform progress reserve is needed for a corresponding sufficiency result.

III. Spatial recovery and minimum entry time

Assume (3), but not strict rest authority. Classical phase-plane/path methods already treat zero initial speed [12], [13], [14], [15]. We also resolve absent launch authority, nonphysical branches, and asymptotic trapping in the viable domain.

For $z_0 = (d_0, v_0) \in D$, use travel $s = d_0 - d$, energy $E = v^2/2$, and $S_0 = d_0 - \epsilon$. For $s < S_0$, define

$$B(s) = \frac{1}{2} \min\{(d_0 - s - \epsilon)/\gamma, M\}^2,$$
$$F(s, E) = A_+(d_0 - s, \sqrt{2E}). \tag{10}$$

If $d_0 = \epsilon$, or $v_0 = 0$ and $A_+(d_0, 0) = 0$, set $s_*(z_0) = 0$. Otherwise construct the upper-constrained profile $E^+(0) = v_0^2/2$ by

$$(E^+)' = \begin{cases} F(s, E^+), & E^+ < B(s), \\ \min\{F(s, E^+), B'(s)\}, & E^+ = B(s), \end{cases} \tag{11}$$

until its first post-departure zero or S_0, denoted s_*; derivatives and switching hold almost everywhere. Positive-authority launch gives $E^+(s) = A_+(d_0, 0)s + o(s)$; artificial restarts are excluded. Whole-domain viability ensures attainable braking at ceiling contact.

Theorem 3 (Recoverability and entry value)**.** *The profile is well defined and physically realized by a safe, measurable admissible input, with*

$$d(t) \longrightarrow d_0 - s_*, \qquad v(t) \longrightarrow 0.$$

Every safe trajectory has limiting gap at least $d_0 - s_*$*; thus the exact asymptotic recoverability set is*

$$\mathcal{C}_{\text{asy}} = \{(d_0, v_0) \in D : d_0 - s_*(d_0, v_0) \leq d_g\}. \tag{12}$$

For $\eta > 0$*, let* $S_\eta = d_0 - d_g - \eta$*. The minimum first-entry time into* $d \leq d_g + \eta$*, over all safe measurable inputs, is*

$$T_\eta^*(z_0) = \begin{cases} 0, & S_\eta \leq 0, \\ \displaystyle\int_0^{S_\eta} \frac{ds}{\sqrt{2E^+(s)}}, & 0 < S_\eta < s_*, \\ +\infty, & S_\eta > 0,\ S_\eta \geq s_*. \end{cases} \tag{13}$$

Every finite positive value is attained by the physical realization of (11)*.*

Proof. The 2-Lipschitz one-sided Skorokhod map [16] gives local contraction at positive energy; face tests put the slope in $[A_-, A_+]$. Physical full-throttle launch gives $E^+(s) = A_+(d_0, 0)s + o(s)$, with uniqueness from the integrable comparison coefficient $C/\sqrt{s}$. Bounded slopes allow continuation to zero energy or the corner.

For any safe time trajectory, bounded acceleration gives $|E(t_2) - E(t_1)| \leq C[s(t_2) - s(t_1)]$ for $t_2 \geq t_1$. Hence a Lipschitz spatial profile e exists through rests, with $e \leq B$, $e' \leq F(s, e)$ a.e. On $e > E^+$ the upper profile is below its ceiling, so positive-part comparison gives $e \leq E^+$. Change of variables with truncated integrands gives $\int_0^r (2e)^{-1/2} ds \leq T(r)$, where $T(r)$ is elapsed time to distance r. Waiting can only increase this time.

Clock inversion and $u = ((E^+)' - a)/b$ realize E^+. At an interior terminal zero, $A_+(d_0 - s_*, 0) = 0$; otherwise energy would increase nearby. Local Lipschitz full-throttle dynamics exclude finite-time arrival at this equilibrium. At the corner, $E^+(s) \le (S_0 - s)^2/(2\gamma^2)$ gives logarithmic clock divergence. No competitor can travel beyond s_* in finite time. The no-escape argument of Theorem 2 handles nonlaunchable states. The clock and limiting-gap assertions follow; expanded arguments are in [26, Lemmas 6.1–6.2]. □

The value concerns free-terminal-speed gap entry, not minimum-time arrival at $\mathcal{T}_g$. For $h_g = 0$, safety gives $d(t) - \epsilon \ge (d_0 - \epsilon)e^{-t/\gamma}$, excluding finite corner arrival. The optimal input need not be regular. Computing the benchmark requires one scalar constrained ODE and quadrature per initial state; discretization error is separate from the exact value.

Viable general ceilings, partial recoverability, and the interior HJB interpretation appear in [26, Prop. 6.4, Secs. 6.5–6.7]; no global continuous-viscosity uniqueness is claimed. A particular filter need not preserve the system's full recoverability set.

IV. Reflected filtering and fidelity cost

Set $d_\eta = d_g + \eta$ and fix $S = d_0 - d_\eta \in (0, s_*)$. Compute E^+ once for this episode and set

$$f_\pm(s, E) = A_\pm(d_0 - s, \sqrt{2E}),$$
$$q(s) = \theta E^+(s), \qquad 0 < \theta < 1.$$

Require the additional floor condition

$$q' \le f_+(s, q) \quad \text{a.e. on } [0, S]. \tag{14}$$

The face tests concern the ceiling and do not imply this floor test.

Let $k(t, d, v) \in [-U, U]$ be jointly locally Lipschitz and write $p_N = a + bk$, evaluated at $(t, d, v) = (\tau(s), d_0 - s, \sqrt{2E})$. An absolutely continuous profile $E : [0, S] \to \mathbb{R}$ in $q \le E \le B$ specifies the controller through Skorokhod reflection, almost everywhere [16]:

$$E' = \begin{cases} p_N, & q < E < B, \\ \max\{p_N, q'\}, & E = q, \\ \min\{p_N, B'\}, & E = B, \end{cases} \tag{15}$$

$$\tau(s) = \int_0^s (2E(r))^{-1/2}\, dr, \qquad E(0) = v_0^2/2. \tag{16}$$

Invert the clock and apply $u = (E' - a)/b$. At a rest start use the no-waiting branch, whose initial acceleration is $\max\{a(d_0, 0) + b(d_0, 0)k(0, d_0, 0), \theta A_+(d_0, 0)\} > 0$. An active floor arc is constrained sliding motion with $E' = q'$. The energy inequality alone permits waiting, since its time derivative is $v(p - q')$. The episode profile is never reset. Denote the first physical entry time of this reflected controller under nominal k by $T_\theta[k] = \inf\{t \ge 0 : d(t) \le d_\eta\} = \tau(S)$.

Theorem 4 (Competitive entry bound)**.** *Under the original face certificate and* (14)*,* (15)–(16) *has a unique reflected energy profile and a safe physical realization with measurable admissible input. Uniformly over the nominal class,*

$$T^* \le T_\theta[k] \le T^*/\sqrt{\theta}. \tag{17}$$

The nominal is applied whenever it meets the active tangent conditions; otherwise the input is its nearest feasible scalar projection.

Proof. At positive energy, 2-Lipschitz one-sided reflection gives local contraction for trajectory and clock. Separation and bounded slopes allow only one boundary on short intervals; Lipschitz boundaries permit continuation. At rest, $q(s) = \theta g_0 s + o(s)$, $g_0 = A_+(d_0, 0) > 0$. Use the complete trial class $q(s) \le e(s) \le Hs$, $e(0) = 0$, with H sufficiently large; lower reflection preserves this class. The weighted norm $\|\Delta e\|_w = \sup |\Delta e(s)|/s$ gives $|\Delta\tau(s)| \le C\sqrt{s}\|\Delta e\|_w$; nominal Lipschitz continuity gives the same acceleration-difference bound. Integration and lower reflection contract by $C\sqrt{\delta}$ on $[0, \delta]$. The fixed point satisfies $E(s) = p_0 s + o(s)$, $\tau(s) = \sqrt{2s/p_0} + o(\sqrt{s})$, with the stated $p_0 > 0$; inversion gives no-wait launch. Floor and ceiling tests keep $E' \in [f_-, f_+]$. Integrating the clocks in $\theta E^+ \le E \le E^+$ proves (17). □

An admissible episode filter Π assigns a bounded-input physical trajectory to every nominal, including $k = -U$. It is *faithful* if it preserves the band, has $s(t) > 0$ for every $t > 0$ before entry (no initial wait), and applies the nominal strictly above the floor whenever tangent to the original domain. Thus no anticipation above the floor is allowed. Write $T_\Pi[k] = \inf\{t \ge 0 : d_{\Pi[k]}(t) \le d_\eta\}$ for its first-entry time, allowing $+\infty$ before finite entry is proved. Define

$$(E_\theta^-)' = \begin{cases} f_-(s, E_\theta^-), & E_\theta^- > q, \\ \max\{f_-(s, q), q'\}, & E_\theta^- = q, \end{cases} \qquad E_\theta^-(0) = v_0^2/2. \tag{18}$$

Theorem 5 (Exact fidelity cost)**.** *Under the hypotheses of Theorem 4, every admissible faithful episode filter* Π *satisfies*

$$\sup_k T_\Pi[k] = W_\theta := \int_0^S \frac{ds}{\sqrt{2E_\theta^-(s)}} \le T^*/\sqrt{\theta}. \tag{19}$$

The braking nominal $k = -U$ attains W_θ. If $v_0 = 0$ and $f_-(s, q) \le q' \le f_+(s, q)$ a.e. on $[0, S]$, then $W_\theta = T^*/\sqrt{\theta}$. For $v_0 > 0$, $W_\theta < T^*/\sqrt{\theta}$.

Proof. If entry failed, $s(t) \uparrow s_\infty \in (0, S]$; the positive floor there would bound v away from zero, a contradiction. Where $E_\theta^- > e \ge q$, the lower profile obeys f_- and every admissible trajectory obeys $e' \ge f_-(s, e)$. Positive-part comparison gives $e \ge E_\theta^-$; its coefficient $C/\sqrt{q}$ is integrable through launch. For $k = -U$, fidelity forces $e' = f_-(s, e)$ wherever $e > E_\theta^-$, including the original ceiling since $e > E_\theta^- \ge q$ and $f_- \le B'$ there. The opposite comparison gives equality. The clocks prove (19). A fully admissible floor with $E_0 = 0$ solves (18); a moving start has an initial energy buffer strictly above that floor. □

Proposition 6 (Analytic OVM certificate)**.** *For $a = \alpha(V(d) - v)$, $\alpha > 0$, $b = 1$, set $C = \alpha V + U$ and $\bar{v} = \min\{(d - \epsilon)/\gamma, M\}$. If the original face tests and*

$$\alpha\bar{v}(d) \le 2C(d), \qquad C(d) \le 2U \tag{20}$$

hold for every $d \ge \epsilon$, every $\theta \in (0,1)$ gives a fully admissible floor. Thus the sharp rest-start factor is $1/\sqrt{\theta}$.

Proof. Put $r = \sqrt{\theta}$, $v = \sqrt{2E^+}$. The upper margin is at least $(1-r)[(1+r)C - \alpha r v] \ge 0$. The lower margin under full throttle is $2U - (1-r^2)C + \alpha(r-r^2)v \ge 0$, and on the speed cap it is nonnegative since $C \le 2U$. On the headway ceiling it is $2U - C + v(\alpha r - r^2/\gamma)$, concave in r; its endpoint values are nonnegative by (20) and the headway face test. □

All three numerical models satisfy this certificate analytically for all $d \ge \epsilon$ [26, Prop. 7.3]. For C1 the minimum of $2C - \alpha\bar{v}$ occurs at $d = \epsilon + \gamma M = .412$ and exceeds 1.9199; C2/C3 satisfy $2 \inf C > \alpha M$. Their respective upper bounds on C are $5.094, 5.036, 8.9 < 2U = 10$. For $\theta = 1$ use E^+. Acceleration jumps preclude a bounded-jerk guarantee. Interior CBF restrictions need not be strictly faithful; Section V uses an anticipatory held-input rule. Full launch and sharp-clock proofs are in [26, Thms. 7.1–7.2].

V. Sampled safety and recovery

For autonomous dynamics, hold the actuator u constant; sampled nominals are arbitrary in $[-U, U]$. Intersample safety and finite-time reachability are established in [18], [19], [25], [24]. Here the performance reference is the exact clock of Section III.

Keep the episode profile fixed. Choose $S < S_e < s_*$ with $B > 0$ on $[0, S_e]$ and require the floor test (14) on this extended interval. Set $q = \theta E^+$, $0 < \theta < 1$. For a constant actuator value u, let

$$e_u' = a(d_0 - r, \sqrt{2e_u}) + b(d_0 - r, \sqrt{2e_u})u, \qquad e_u(s) = E. \tag{21}$$

At the initial rest state use the physical positive-launch branch; a profile reaching zero later is rejected. Write $p_0(u) = a(d_0, 0) + b(d_0, 0)u$ and $g_0 = A_+(d_0, 0)$. Define the spatial certificate, including the rest condition, by

$$I_\delta(s,E) = \{u \in [-U,U] : q \le e_u \le B \text{ on } [s, s + M\delta], \\ E > 0 \text{ or } p_0(u) \ge \theta g_0 > 0\}. \tag{22}$$

Because $v \le M$, a certified spatial prefix covers at least one physical hold of length δ.

For computable bounds on the extended band, choose

$$\begin{aligned} 0 < g &\le \min(B - q), \\ P &\ge \sup_{q \le E \le B,\ |u| \le U} |a + bu|, \\ K &= P + \max\{\mathrm{Lip}(q), \mathrm{Lip}(B)\}, \\ \delta_0 &= \frac{1}{M}\min\{S_e - S, g/(4K)\}. \end{aligned}$$

The explicit choice $g = (1-\theta)\min B$ is valid; known coefficient bounds also give $\mathrm{Lip}(q) \le \theta \max\{P, \mathrm{Lip}(B)\}$.

For update times $0 = t_0 < t_1 < \cdots$, define

$$T_{\mathrm{entry}} = \inf\{t \ge 0 : d(t) \le d_\eta\}, \\ T_{\mathrm{detected}} = \inf\{t_n : d(t_n) \le d_\eta\},$$

with $\inf \varnothing = +\infty$. The second time uses exact sampled state observations.

Theorem 7 (Sampled safety and recovery)**.** *Under the autonomous face and floor certificates, $I_\delta(s,E)$ is a nonempty closed interval for every pre-entry state $q(s) \le E \le B(s)$ and $0 < \delta \le \delta_0$. Projection of the sampled nominal onto this interval gives a recursively feasible zero-order-hold controller, safe throughout every hold, including launch from rest. This global period is an existence bound. Proposition 8 gives a less conservative sufficient condition, with useful values in Corollary 9.*

More generally, fix a maximum period $\bar{\Delta} > 0$ and $\delta_F = \min\{\bar{\Delta}, \delta_0\}$. At each update, any command certified to preserve the band throughout a hold $\delta_n \in [\delta_F, \bar{\Delta}]$ may be used, provided that its travel stays in $[0, S_e]$ and a rest start has a positive, no-waiting launch. If no longer hold is certified, use δ_F and the guaranteed spatial certificate. The update times cannot accumulate, and

$$\begin{aligned} T^* \le T_{\mathrm{entry}} \le W_\theta \le T^*/\sqrt{\theta}, \\ T_{\mathrm{detected}} \le W_\theta + \bar{\Delta}. \end{aligned} \tag{23}$$

Detection in this statement uses exact state observations at updates. Safety holds through detected entry. A post-entry sampled controller requires its own certificate.

Proof. Scalar comparison orders the profiles in (21) by u, so the feasible set is an interval. Feasible profiles have bounded slope; compactness and their integral equations preserve the band under input limits. The closed rest condition preserves positive launch, proving closedness. If $B(s) - E \ge g/2$, use $u = U$: the floor is a subsolution, while the clearance from the ceiling satisfies $B(r) - e_U(r) \ge g/2 - K(r-s) \ge g/4$ until any first exit. Otherwise use $u = -U$: the ceiling is a supersolution and the analogous lower clearance exceeds $g/4$. Thus no first exit occurs over $M\delta \le g/(4K)$. The held profile's clock obeys $\tau(s + M\delta) = \int_s^{s+M\delta} (2e_u)^{-1/2} dr \ge \delta$; inversion realizes the same constant actuator over a physical hold. The extension condition makes this argument available at every update. The floor and launch rule give finite entry as in Theorem 5. The first comparison in Theorem 5 uses only admissibility and the floor, so $E_\Delta \ge E_\theta^-$ for the sampled trajectory. Its clock is therefore at most $W_\theta \le T^*/\sqrt{\theta}$. The final hold delays detection by at most $\bar{\Delta}$; the lower hold bound excludes Zeno updates. Appending the safe continuous-time backup after detected entry makes this finite trajectory part of a globally safe admissible input, so the definition of T^* also gives $T^* \le T_{\mathrm{entry}}$ [26, Thm. 8.3]. □

The clock bound implies neither pairwise speed improvement nor sampled-class sharpness. Interval structure concerns the common spatial prefix, not the physical-time test [26, Sec. 8.3].

a) A useful a priori sampling period: Under the same episode assumptions, write $\mathsf{F}(r,e,u) = a(d_0 - r, \sqrt{2e}) + b(d_0 - r, \sqrt{2e})u$. Choose a common prefix $J_s = [s, r_s]$ by

$$\int_s^{r_s} \frac{dr}{\sqrt{2B(r)}} = \Delta, \qquad \Delta \le \int_S^{S_e} \frac{dr}{\sqrt{2B(r)}}. \tag{24}$$

Its headway-branch length is $(d_0 - s - \epsilon)(1 - e^{-\Delta/\gamma})$. Choose constant $u_q, u_B \in [-U, U]$ such that $q' \le \mathsf{F}(r, q, u_q)$ and $\mathsf{F}(r, B, u_B) \le B'$ almost everywhere on J_s. At rest require $a(d_0, 0) + b(d_0, 0)u_q \ge \theta g_0 > 0$. For $q(r) \le e \le B(r)$ let nonnegative c_B, c_q bound $\mathsf{F}(r, e, u_q) - B'$ and $q' - \mathsf{F}(r, e, u_B)$, respectively, and put $H_B = \int_{J_s} c_B$, $H_q = \int_{J_s} c_q$.

Proposition 8 (Boundary-clearance certificate)**.** *If, for every $s \in [0, S]$,*

$$H_B + H_q \le B(s) - q(s), \tag{25}$$

the spatial feasible-input interval on J_s is nonempty and closed at every band state. Fixed holds of length Δ are recursively feasible, including launch, and retain $T^* \le T_{\text{entry}} \le W_\theta$ and $T_{\text{detected}} \le W_\theta + \Delta$. A feasible witness is u_q if $B(s) - E \ge H_B$, and u_B otherwise. A nominal may instead pass a sound physical-hold certificate as in Theorem 7; the spatial witness supplies its fallback.

Proof. Scalar comparison preserves the floor under u_q and the ceiling under u_B. In the first branch, upper clearance is at least $B(s) - E - \int_s^r c_B \ge 0$. Equality at an interior first contact leaves zero remaining budget, so the ceiling is a supersolution thereafter. In the other branch, (25) gives $E - q(s) > H_q$, excluding lower exit. Rest selects u_q and its positive physical branch. Ordering and compactness give the interval as in Theorem 7. Each profile's clock is at least the ceiling clock in (24); the extension therefore proves recursion. The floor comparison gives the upper recovery bounds. Appending the safe backup after detected entry places the trajectory in the globally safe input class defining T^*, which proves the lower bound [26, Thm. 8.9]. □

For the OVM, F decreases in energy, so $c_B = [\mathsf{F}(r, q, u_q) - B']_+$ and $c_q = [q' - \mathsf{F}(r, B, u_B)]_+$ suffice. The general bound follows by choosing $u_q = U$, $u_B = -U$.

Corollary 9 (Periods for specified OVM bands)**.** *For the three models in Table I, $h_0 = .8$, $v_0 \in \{0, .4\}$, both targets, and $\theta \in \{.25, .5\}$, with extension $d - \epsilon = .002$, the common period $\Delta = .01$ admits an explicit analytic witness throughout recovery. An offline interval certificate further establishes $\Delta = .025$. Both values concern every band state and arbitrary sampled nominals for six fixed initial-state profiles, two θ values and two targets (12 bands, 24 combinations), in normalized model time. The analytic construction uses $\alpha\gamma \le 1$ and $C > 0$ on the extended episode. Every accepted rest-start input must satisfy the positive launch guard.*

Proof. Write $x = d - \epsilon$, $C = \alpha\mathcal{V} + U$ and $v_{\text{cap}} = \min(M, x/\gamma)$. The following bounds hold on the extended episodes; c_T bounds C below for $x \ge x_T$:

Model	$C_{\min}$	$C_{\max}$	c_T	x_T
C1	2.55	4.36	3.05	.20
C2	4.05	4.78	4.43	.40
C3	5.00	6.79	5.76	.45

Put $v_T = x_T/\gamma$. A constant-authority reflected lower profile reaches the ceiling by x_T: its rest-start distance to v_T is

$$D_T = \frac{c_T}{\alpha^2} \log \frac{c_T}{c_T - \alpha v_T} - \frac{v_T}{\alpha}, \tag{26}$$

bounded above by $.479, .253, .246$, respectively, while available distances are $.60, .40, .35$. A moving start improves both inequalities. Then $\alpha\gamma \le 1$ and $C > 0$ keep $E^+ = B$ for $x \le x_T$.

For $x \ge x_T$, certify the longer prefix of length $v_{\text{cap}}\Delta$ using $U, -U$. Their opposite-clearance rates are bounded by $C_{\max} + v_{\text{cap}}/\gamma$ and $[2U - (1-\theta)C_{\min}]_+ + \alpha v_{\text{cap}}$. Since $B - q \ge (1-\theta)v_{\text{cap}}^2/2$, it is sufficient that

$$\Delta \le \frac{(1-\theta)v_T}{2\{C_{\max} + [2U - (1-\theta)C_{\min}]_+ + (\alpha + 1/\gamma)v_T\}}. \tag{27}$$

For $x < x_T$, set $\kappa_q = \alpha\sqrt{\theta}/\gamma - \theta/\gamma^2$, $\kappa_B = \alpha/\gamma - 1/\gamma^2$, and $\Lambda_j = \alpha L_{\mathcal{V}} + |\kappa_j|$, where $L_{\mathcal{V}} = 1, 1, .8$ for C1–C3, respectively. With $\ell = x(1 - e^{-\Delta/\gamma})$, feasible boundary witnesses are

$$u_q = \min\{U, -\alpha\mathcal{V} + \kappa_q x + (\alpha L_{\mathcal{V}} - \kappa_q)_+ \ell\},$$
$$u_B = -\alpha\mathcal{V} + \kappa_B x.$$

Monotonicity of $\mathcal{V}$ makes them preserve their respective boundaries. Their opposite-clearance rates are at most $(1-\theta)x/\gamma^2 + \Lambda_j \ell$. Thus, putting $z = \ell/x$, the sufficient tail condition is

$$2z + \frac{\gamma^2(\Lambda_q + \Lambda_B)}{1-\theta} z^2 \le \tfrac{1}{2}. \tag{28}$$

The smallest early bound is greater than .01208; the smallest tail bound is greater than .04786. The extension condition $\Delta \le \gamma(1 - .002/.01)$ is at least .16. Hence .01 works throughout every episode, including a full-throttle rest launch.

For .025, outward constant-authority profile enclosures and nonnegative budget sums verify (25) on 101,600 complete sample-location cells of width 10^{-4}, with maximum upper budget/width ratio below .748. Cells containing profile contact enclose both derivative branches. Enlarged prefixes cover every starting position in each cell, and monotonicity in energy covers the whole band. This is an offline uniform certificate; it is distinct from the trajectory-specific .05 tests and does not certify a runtime deadline. Full bounds and reproducible certificates are given in [26, Thm. 8.12, Sec. 8.9]. For fixed plant and episode constants, the combined analytic period bound is proportional to $1 - \theta$ as $\theta \uparrow 1$; the pre-tail term eventually determines that scaling. □

b) Validated physical-time holds: For other periods, fix a gap interval K_d and verified Lipschitz bound $L_{\mathcal{V}}$. A numerical

path $\widehat{z} = (\widehat{d}, \widehat{v})$ has defects $r_d = \dot{\widehat{d}} + \widehat{v}$ and $r_v = \dot{\widehat{v}} - \alpha(\mathcal{V}(\widehat{d}) - \widehat{v}) - u$. Bounds $\rho \geq |r|$ define error radii $\xi = (\xi_d, \xi_v)^\top$ by

$$\dot{\xi} = H\xi + \rho, \qquad H = \begin{pmatrix} 0 & 1 \\ \alpha L_{\mathcal{V}} & -\alpha \end{pmatrix}. \tag{29}$$

Initialize $\xi(0) = \xi_0 \geq |z_0 - \widehat{z}_0|$ componentwise and verify $[\widehat{d} - \xi_d, \widehat{d} + \xi_d] \subset \operatorname{int} K_d$ throughout the hold. For $e = z - \widehat{z}$, $D^+|e| \leq H|e| + \rho$ componentwise. Positivity of e^{Ht} and the tube inclusion give $|e| \leq \xi$ by comparison and continuation. At an approximation jump, propagate $\xi^+ \geq \xi^- + |\widehat{z}^+ - \widehat{z}^-|$. Let $Q \geq E^+$ be a verified upper reference and put $J = [d_0 - \widehat{d} - \xi_d, d_0 - \widehat{d} + \xi_d]$. The sufficient whole-hold acceptance tests are

$$\begin{aligned} &\widehat{v} - \xi_v \geq 0, \qquad \widehat{v} + \xi_v \leq M, \\ &\widehat{d} - \gamma\widehat{v} - \epsilon \geq \xi_d + \gamma\xi_v, \\ &\tfrac{1}{2}(\widehat{v} - \xi_v)^2 \geq \theta \sup_{r \in J} Q(r), \qquad J \subset [0, S_e]. \end{aligned} \tag{30}$$

They must hold throughout every cell, including propagated initial error and outward arithmetic. A full-throttle comparison supplies an analytic floor certificate at rest, where a strictly separated numerical floor test would be inappropriate. Safety and travel-domain checks are still required.

For nondecreasing $\mathcal{V}$, write $C(s) = \alpha\mathcal{V}(d_0 - s) + U$. On a spatial cell $[s_j, s_{j+1}]$, choose $C_j \geq C(s_j)$, $C_j > 0$, and $0 < \underline{\alpha} \leq \alpha$, with verified bounds. Let $\overline{\Phi}_j(h; e)$ enclose the flow of $e' = C_j - \underline{\alpha}\sqrt{2e}$ from above. With $\overline{E}_0 \geq v_0^2/2$ and $h_j = s_{j+1} - s_j$, set

$$\overline{E}_{j+1} = \min\{\overline{B}(s_{j+1}), \overline{\Phi}_j(h_j; \overline{E}_j)\}. \tag{31}$$

Since $(E^+)' \leq C_j - \underline{\alpha}\sqrt{2E^+}$, scalar comparison proves the endpoint bounds inductively. An online bound is

$$Q(s) = \min\{\overline{B}(s), \overline{E}_j + C_j(s - s_j)\}, \tag{32}$$

with upward evaluation and $\overline{B} \geq B$. The flow has an elementary physical clock; verified root brackets enclose its value. For $s_j < S$, put $h_j = \min\{S, s_{j+1}\} - s_j$, $x_j = d_0 - s_j - \epsilon$, and

$$\begin{aligned} \ell_j &= \frac{2h_j}{\sqrt{2(\overline{E}_j + C_j h_j)} + \sqrt{2\overline{E}_j}}, \\ \underline{T}_Q &= \sum_j \max\{\ell_j, h_j/M, \gamma \log[x_j/(x_j - h_j)]\} \leq T^*. \end{aligned} \tag{33}$$

Integrated affine-energy, speed, and headway bounds give this lower clock. Downward evaluation of $\underline{T}_Q/\sqrt{\theta} + \bar{\Delta}$ and an upper observed time certify the simpler bound, not W_θ [26, Lemmas 8.4–8.6]. Conservative interval observation differs from exact sampled observation. Propagated uncertainty may require refinement: the theorem does not guarantee a coarse finite search terminates. Testing the nominal first preserves every accepted hold, but anticipates boundary contact; strict fidelity, its exact adversarial value, and minimum integrated intervention are not asserted for this sampled rule.

For a sampled-safe continuation in the running model, suppose $\alpha\gamma \leq 1$, $\mathcal{V}$ is nondecreasing, $L_{\mathcal{V}}$ also bounds its Lipschitz constant on $[\epsilon, \epsilon + \gamma M]$, and detected entry has $d \leq \epsilon + \gamma M$. The held command

$$u_n = -\alpha\mathcal{V}(d_n) + (\alpha - 1/\gamma)v_n \tag{34}$$

preserves D for every fixed tail period

$$0 < \Delta < \Delta_{\text{tail}} := \begin{cases} \arctan(\gamma\sqrt{\alpha L_{\mathcal{V}}})/\sqrt{\alpha L_{\mathcal{V}}}, & L_{\mathcal{V}} > 0, \\ \gamma, & L_{\mathcal{V}} = 0. \end{cases} \tag{35}$$

At $v_n = 0$ the state stays at rest. Otherwise $p(0) = -v_n/\gamma$, $\dot{p} = -\alpha p - \alpha m(t)v$ with $0 \leq m \leq L_{\mathcal{V}}$. Hence $p(t) \leq p(0)e^{-\alpha t} < 0$ while $v > 0$, and $r = -p/v$ satisfies $\dot{r} \leq r^2 + \alpha L_{\mathcal{V}}$. This keeps $v > 0$ through the hold; $\dot{h} = (\alpha\gamma - 1)p + \alpha\gamma m v \geq 0$ gives safety. The command decreases with v; rest and headway face tests bound its endpoints by U and $-U$. Moreover $v_{n+1} \leq [1 - (1 - e^{-\alpha\Delta})/(\alpha\gamma)]v_n$. The factor lies in $(0, 1)$ since $\Delta < \Delta_{\text{tail}} \leq \gamma$. Thus $v \to 0$ and $d_\infty \in [\epsilon, d_{\text{detected}}]$ [26, Prop. 8.7]. This preserves the tolerance target, and does not assert exact arrival at the smaller rest target $\mathcal{T}_g$.

TABLE I
OPTIMAL-VELOCITY MODELS.

Model	α	γ	$\mathcal{V}(d)$
C1	2.6	.20	$\tanh d - \tanh 2$
C2	1.0	.30	$\tanh d - \tanh 2$
C3	2.6	.35	$\text{clip}_{[0,1.5]}(.8(d - .1))$

VI. Certification and numerical comparisons

The normalized models in Table I use $a = \alpha(\mathcal{V}(d) - v)$, $b = 1$, $\epsilon = .02$, $M = 1.96$, $U = 5$. They satisfy the face and all-θ floor certificates and test structure; they are not calibrated vehicles.

A. Protocol and controllers

The gap task uses $h_g = 0$, $\ell = .2u^2 + 50d^2$; the tracking-cost task uses $h_g = .2$, $\ell = .2u^2 + v^2 + 5(h - .2)^2$. Both seek free-terminal-speed entry into $d \leq d_\eta = \epsilon + h_g + .01$; tracking names the cost. Initial states have $h_0 = .8$, $v_0 \in \{0, .4\}$, and

$$d_0 = \epsilon + h_0 + \gamma v_0. \tag{36}$$

The four nominals are a finite-horizon HJB policy, $k = 0$, clipped drift cancellation, and braking $k = -U$. HJB specifications and regeneration checks are in [26, Sec. 13.1.1].

Each period uses 48 model/task/state/nominal combinations at $\theta = .25, .5$. The .05 comparison tests band filtering, naive ZOH of the reflected law, and a whole-hold-certified progress CBF. The progress request tracks $\sqrt{\theta}\min\{M, (d - \epsilon)/\gamma\}$ with gain 5 and safety gains 10. It uses the same safety certificate, without the floor. This structure-matched CLF/CBF baseline is not a reproduction of a published sampled-CBF implementation.

The .05 band search tests nominal/grid/seed candidates, refines toward the request and permits halving. Full rules and baseline formulas are in [26, Secs. 13.2–13.5]. Offline profile enclosures and whole-cell residuals certify holds without state projection [29]; rejection does not prove physical infeasibility.

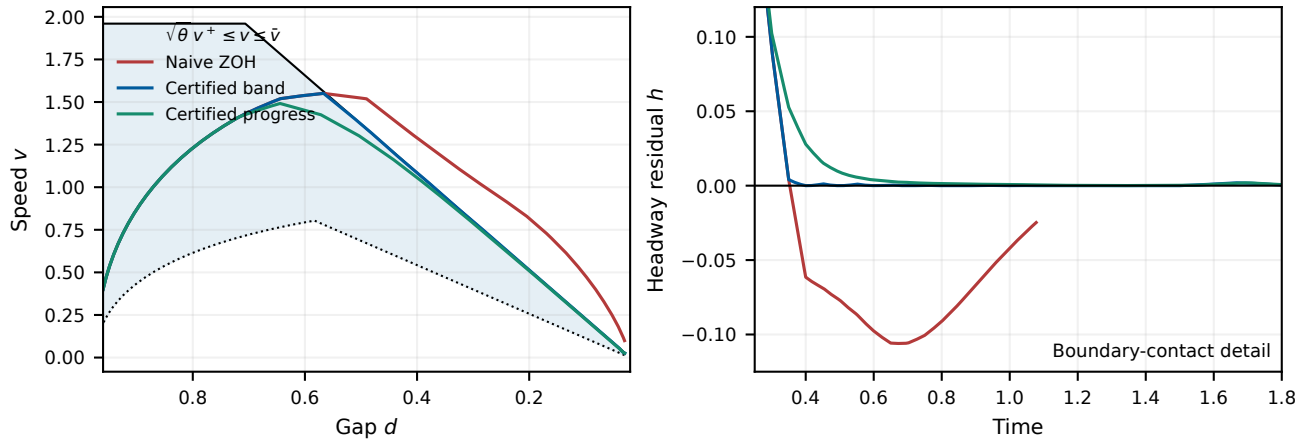


Fig. 1. C3 moving-start gap task, HJB nominal, $\theta = .25$, $\bar{\Delta} = .05$. Speed–gap (left) and headway (right) diagnostics: naive ZOH leaves the safe domain; both whole-hold certified methods remain safe. The band is shown in speed coordinates.

TABLE II

MAIN MATRIX AT REQUESTED PERIOD .05. DOMAIN/FLOOR EXITS ARE DIAGNOSTIC FOR NAIVE ZOH; CERTIFIED METHODS ENFORCE WHOLE-HOLD SAFETY. †: FLOOR DEPARTURES ARE DIAGNOSTIC, SINCE PROGRESS DOES NOT ENFORCE THE FLOOR.

Controller	N	Entry	Domain exits	Floor exits
Naive ZOH	96	70	60	88
Certified band	96	96	0	0
Certified progress	96	96	0	55†

TABLE III

ENTRY RATIOS AT .05: 48 CASES PER ROW; PROGRESS USES $r = \sqrt{\theta}$.

Method	θ	Median T/T^*	Maximum T/T^*
Band	.25	1.745	1.929
Progress	.25	1.467	1.894
Band	.50	1.303	1.391
Progress	.50	1.130	1.383

TABLE IV

PAIRED COMMON-HORIZON COMPARISONS AT .05; 96 MAIN PAIRS.

Metric	Band lower	Progress lower
Entry time	18/96	78/96
J_{10}	18/96	78/96
$I_{1,10}$	65/96	31/96
$I_{2,10}$	25/96	71/96
Acceleration RMS	24/96	72/96
Acceleration TV	1/96	95/96

B. Fixed-period certificate exercised

A separate implementation accepts a certified nominal hold or applies the stored Proposition 8 witness. It carries an outward state enclosure, accounts for decimal-parameter rounding and checks the rest guard. All 192 runs at fixed periods .01 and .025 enter with no shortened recovery hold, failed certificate or uncertainty-blocked witness. The maximum diagnostic $T_{\text{entry}}/W_\theta$ is below .990619 and .980558, respectively; all conservative entry-bracket checks also pass. Figure 2 reports these experiments. The certificates include the floor kink.

C. Safety and performance comparisons

The .05 comparison and .0125/.025/.1 stress sweep contain 360 runs. All 120 band and 120 progress runs enter and complete the ten-unit horizon. All 96 main band runs retain .05; one .1 stress run needs a .05 hold. These longer-hold certificates are trajectory-specific. The generic existence bound is 5.84×10^{-8}–2.14×10^{-7}; Corollary 9 instead supplies the uniform .01/.025 episode-band guarantees exercised above.

Naive ZOH reaches headway $-.11035$ (Fig. 1); the minimum certified main headway is 4.12×10^{-10}. Every certified run passes the observer check $\overline{T}_{\text{obs}} \le \underline{T}^*/\sqrt{\theta} + \bar{\Delta}$, using a verified lower clock and upper gap enclosure. Progress meets the tested bound despite 55 floor departures: floor invariance is sufficient, not necessary for an individual trajectory's bound.

After confirmed entry, certified controllers share the sampled stopping continuation. Common-horizon metrics are cost, actual-nominal absolute and squared intervention, acceleration RMS and variation including hold jumps. These imply neither bounded jerk nor calibrated comfort.

Progress is faster and cheaper separately in 78/96 pairs, jointly in 66/96. Table IV also retains unfavorable motion outcomes. These deterministic cases are not independent samples. The witness policy enters earlier than the .05 search in 21/96 and 28/96 cases at .01/.025. Both period and selection rule change; these comparisons imply no general speed ordering.

Latency is omitted from the plant model. The .05 search's serialized median is 69.2 ms versus 2.50 ms for progress. The witness policy's serial medians are 5.94/8.09 ms, with maxima 45.25/44.33 ms, at .01/.025 over 16 stress episodes. These measurements establish no hard deadline. Full records

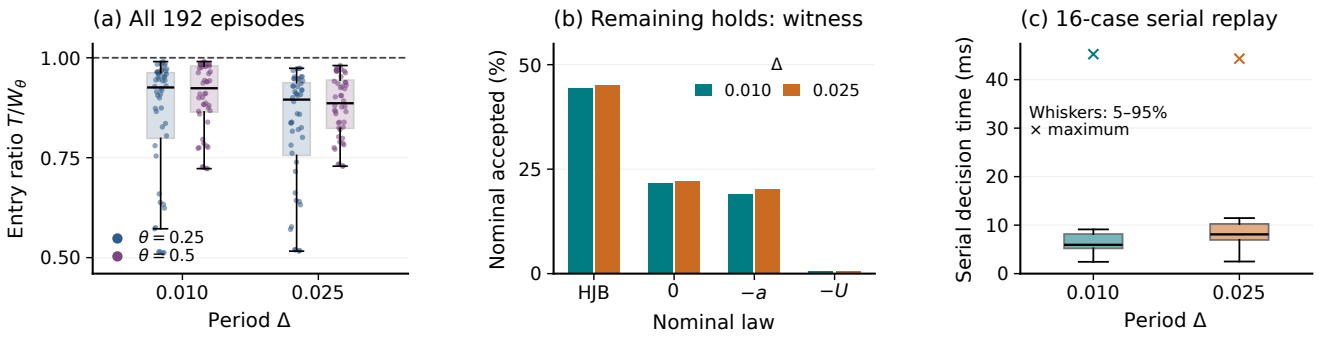


Fig. 2. Nominal-first filtering with the stored clearance witness: 96 cases at each certified fixed period. Entry ratios are trajectory diagnostics; band invariance uses whole-hold certificates. Acceptance counts and serialized decision times refer to the implemented policy.

distinguish the preserved .05 study from the new runs [26, Sec. 13].

VII. Conclusion

The exact fidelity clock W_θ and constructive boundary-clearance certificate provide joint safety/recovery guarantees with useful held-input periods, including rest launch. Exact performance uses autonomous stopped-leader dynamics and positive-tolerance entry with free terminal speed. Matched comparisons quantify intervention, performance, and computation; the certificate implies no pairwise speed, cost, or runtime dominance.

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